

WordPiece Tokenizer — Worked Math Example

Corpus: 3 words — ab (freq 10), ac (freq 1), bc (freq 1)

Setup

Initial alphabet: a, b, ##b, ##c

Initial splits:

word	split
ab	[a, ##b]
ac	[a, ##c]
bc	[b, ##c]

$$\text{score}(x, y) = \text{freq}(x, y) / (\text{freq}(x) \times \text{freq}(y))$$

Round 1

Letter frequencies:

symbol	freq
a	$10 + 1 = 11$
##b	10
##c	$1 + 1 = 2$
b	1

Pair frequencies and scores:

pair	freq	score
(a, ##b)	10	$10 / (11 \times 10) = 0.0909$
(a, ##c)	1	$1 / (11 \times 2) = 0.0455$
(b, ##c)	1	$1 / (1 \times 2) = 0.5 \leftarrow \text{highest}$

Merge: b + ##c → bc

word	split
ab	[a, ##b]
ac	[a, ##c]
bc	[bc]

Vocab so far (5): a, b, ##b, ##c, bc

Round 2

bc is now one symbol — contributes only a letter count, no pair.

Letter frequencies:

symbol	freq
a	$10 + 1 = 11$
##b	10
##c	1
bc	1

Pair frequencies and scores:

pair	freq	score
(a, ##b)	10	$10 / (11 \times 10) = 0.0909$
(a, ##c)	1	$1 / (11 \times 1) = 0.0909$

Tie → break using higher raw frequency: (a, ##b) wins.

Merge: a + ##b → ab

word	split
ab	[ab]
ac	[a, ##c]
bc	[bc]

Vocab so far (6): a, b, ##b, ##c, bc, ab

Round 3

Letter frequencies:

symbol	freq
ab	10
a	1
##c	1
bc	1

Only one pair remains:

pair	freq	score
(a, ##c)	1	$1 / (1 \times 1) = 1.0$

Merge: a + ##c → ac

word	split
ab	[ab]

ac	[ac]
bc	[bc]

Every word is now a single symbol — no pairs left, training stops.

Final Result

Vocabulary (7 tokens): a, b, ##b, ##c, bc, ab, ac

Merge rules learned, in order:

1. $b + \text{\#}\text{\#}c \rightarrow bc$
2. $a + \text{\#}\text{\#}b \rightarrow ab$
3. $a + \text{\#}\text{\#}c \rightarrow ac$